with supernatural powers. Seven is a "lucky" number. In the New Testament's Book of Revelations, seven seals on a scroll are opened, seven trumpets are sounded, seven bowls are filled. St. Augustine obscurely argued for the mystic importance of seven on the grounds that three "is the first whole number that is odd" (what about one?), "four the first that is even" (what about two?), and "of these … seven is composed." And so on. Even in our time these associations linger.

The existence even of the four satellites of Jupiter that Galileo discovered—hardly planets—was disbelieved on the grounds that it challenged the precedence of the number seven. As acceptance of the Copernican system grew, the Earth was added to the list of planets, and the Sun and Moon were removed. Thus, there seemed to be only six planets (Mercury, Venus, Earth, Mars, Jupiter, and Saturn). So learned academic arguments were invented showing why there *had* to be six. For example, six is the first "perfect" number, equal to the sum of its divisors (1 + 2 + 3). Q.E.D. And anyway, there were only six days of creation, not seven. People found ways to accommodate from seven planets to six.

As those adept at numerological mysticism adjusted to the Copernican system, this self-indulgent mode of thinking spilled over from planets to moons. The Earth had one moon; Jupiter had the four Galilean moons. That made five. Clearly one was missing. (Don't forget: Six is the first perfect number.) When Huygens discovered Titan in 1655, he and many others convinced themselves that it was the last: Six planets, six moons, and God's in His Heaven.

The historian of science I. Bernard Cohen of Harvard University has pointed out that Huygens actually gave up searching for other moons because it was apparent, from such arguments, that no more were to be found. Sixteen years later, ironically with Huygens in attendance, G. D. Cassini* of the Paris Observatory discovered a seventh moon—Iapetus, a bizarre world with one hemisphere black and the other white, in an orbit exterior to Titan's. Shortly after, Cassini discovered Rhea, the next Saturnian moon interior to Titan.

Here was another opportunity for numerology, this time harnessed to the practical task of flattering patrons. Cassini added